

Math 340 Midterm 2 Information and Review

- Midterm 2 date, time, and room information are posted on Canvas. It will cover Chapters 12-14, 16-22, 24-26 in your textbook.
- Midterm 2 is a 90-minute exam consisting of True/False questions, multiple choice problems, and free response questions.
- The only permitted materials are: pencil/pen. Any other materials/resources are prohibited.
- This review is intended to give you extra practice with connecting the concepts covered in these sections. The problems on the exam are not guaranteed to be similar to these problems.
- For additional practice, I recommend going through your homework problems, discussion worksheets, and lecture examples.

Terms to know: directed line segment, initial point, terminal point, components, scalar, vector, n -dimensional vector, n -dimensional space, zero vector, vector addition, negative (of a vector), vector subtraction, scalar multiple of a vector, parallel vector, linear combination, standard basis of $\mathbb{R}^n$, norm/length/magnitude, unit vector, normalization, distance, dot product, angle, orthogonal vectors, orthogonal projection, vector component of a vector $\mathbf{u}$ orthogonal to a second vector, vector addition, scalar multiplication, closure/closed under addition/scalar multiplication, vector space, vector, zero vector, negative vector, vector subtraction, trivial vector space, $\mathbb{R}^n$, $M_{m \times n}$, P , P_d , subspace, trivial subspace, linear combination, subspace generated by a set of vectors, spanning set/generating set, solution space of a homogeneous system, linearly dependent and linearly independent, basis, ordered basis, coordinates of a vector relative to an ordered basis B , coordinate vector associated to a vector $\mathbf{w}$ relative to a basis B , finite-dimensional vector space, infinite-dimensional vector space, dimension of a finite-dimensional vector space, column Space, Row Space, Null Space, Nullity, Rank, transition matrix/change of basis matrix, eigenvector, eigenvalue, eigenspace, characteristic polynomial, characteristic equation, Similar matrices, transition matrix, diagonalizable, algebraic multiplicity, geometric multiplicity, similar matrices, trace

Learning Goals for Midterm 2

Clicking on each section will refer you to corresponding page in the review packet.

- Be able to compute the dot product of vectors
- Be able to compute length of vectors in $\mathbb{R}^n$ and find a unit vector parallel to a given vector
- Be familiar with the basic properties of the dot product (including the test for orthogonality)
- Be familiar with the Cauchy-Bunyakovsky-Schwarz inequality and its implications
- Know how to compute the orthogonal projection and what it geometrically represents
- If given one of the 10 vector space properties, be able to verify if the property holds/does not hold for a potential vector space
- Be able to come up with a counterexample to show a set/operation is NOT a vector space
- Be able to verify that a subset of a vector space is or is NOT a subspace (i.e. use the Subspace test)
- Be able to determine if a vector is in the span of a set of vectors
- Be able to determine if a set spans a vector space
- Be able to determine if two spanning sets generate the same subspace
- Be able to determine if a set is linearly independent/linearly dependent (via systems of equations, the determinant of a matrix, or demonstrating a linear combination)
- Know how to show a set is or is NOT a basis for a vector space
- Know how to find a basis from a set of vectors
- Be able to find the coordinates of a vector with respect to an ordered basis
- Be able to find the dimension of a vector space/subspace
- Know the dimensions of the common finite-dimensional vector spaces: $\mathbb{R}^n$, P_d , $M_{m \times n}$
- Know how to use the dimension of a vector space to justify if a set is a basis
- Be able to compute the dimension of the row space and the dimension of the column space of a matrix
- Be able to find a basis for the row space, column space/image space, and nullspace of a matrix
- Be able to find the rank and nullity for a matrix, and how the nullity and rank relate to the matrix
- Know how to compute the change matrix/transition matrix given two ordered bases for a vector space, and how to use it to find the coordinate vector
- Know the basic properties of change matrices/transition matrices (composition of transition/change matrices, inversion of transition/change matrices)
- Know how to verify if a vector is an eigenvector for a matrix

- Be able to compute the eigenvectors/eigenvalues (real eigenvalues only) for a matrix using the characteristic equation, and find a basis for the eigenspace
- Know how to use similarity to find shared properties of similar matrices
- Be able to find the algebraic multiplicity and geometric multiplicity of an eigenvalue
- Be able to determine if a matrix is diagonalizable
- Be able to diagonalize a matrix (i.e. find P and D)
- Use a diagonalization to compute higher powers of a matrix
- *** Be familiar with the (evergrowing) list of invertibility criteria (see Box Of Facts)

Explanations and Example Problems

We can expand on the important concepts mentioned in the review packet as follows.

- Be able to compute the dot product of vectors

Definition: Given any two vectors $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$ in $\mathbb{R}^n$, the **dot product** between two vectors $\mathbf{v}$ and $\mathbf{w}$ is defined as

$$\mathbf{v} \cdot \mathbf{w} := v_1 w_1 + v_2 w_2 + \cdots + v_n w_n. \quad (1)$$

- Be able to compute length of vectors in $\mathbb{R}^n$ and find a unit vector parallel to a given vector

Definition: Recall that given a vector $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ in $\mathbb{R}^n$, the length of $\mathbf{v}$ is defined as

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}. \quad (2)$$

Example: The length of the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ is

$$\|\mathbf{v}\| = \sqrt{1^2 + 4^2 + 5^2} = \sqrt{42}. \quad (3)$$

A unit vector in the direction of $\mathbf{v}$ is $\frac{1}{\|\mathbf{v}\|} \mathbf{v} = \frac{1}{\sqrt{42}} \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$

- Be familiar with the basic properties of the dot product (including the test for orthogonality)

[Basic Properties]: The dot product on $\mathbb{R}^n$ has the following properties. (Given $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$).

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
2. $\mathbf{w} \cdot (\mathbf{u} + \mathbf{v}) = \mathbf{w} \cdot \mathbf{u} + \mathbf{w} \cdot \mathbf{v}$.
3. $c\mathbf{u} \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v})$ for any real scalar c .
4. $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}$
5. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.
6. $\|c\mathbf{u}\| = |c|\|\mathbf{u}\|$

Definition: Two vectors $\mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ are orthogonal if and only if $\mathbf{v} \cdot \mathbf{w} = 0$. In particular, for vectors $\mathbf{v}, \mathbf{w}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$, the angle between two vectors θ can be computed from using the relation

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \|\mathbf{w}\| \cos \theta. \quad (4)$$

Example: The set of vectors $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ in $\mathbb{R}^3$ orthogonal to the vector $\mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}$ is the set

$$S = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \text{ in } \mathbb{R}^3 \mid v_1 + 3v_2 - 5v_3 = 0 \right\} \quad (5)$$

In fact, one can prove that S is a 2-dimensional subspace of $\mathbb{R}^3$.

- Be familiar with the Cauchy-Schwarz inequality and its implications

[[**Cauchy-Bunyakovsky-Schwarz**]]: If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in $\mathbb{R}^n$, then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|. \quad (6)$$

[[**Consequence 1: Angle**]] We can define the angle θ between two nonzero vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ as

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad (7)$$

because the Cauchy-Schwarz inequality implies

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1. \quad (8)$$

[[**Consequence 2: Triangle Inequality**]] For any two vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ we have

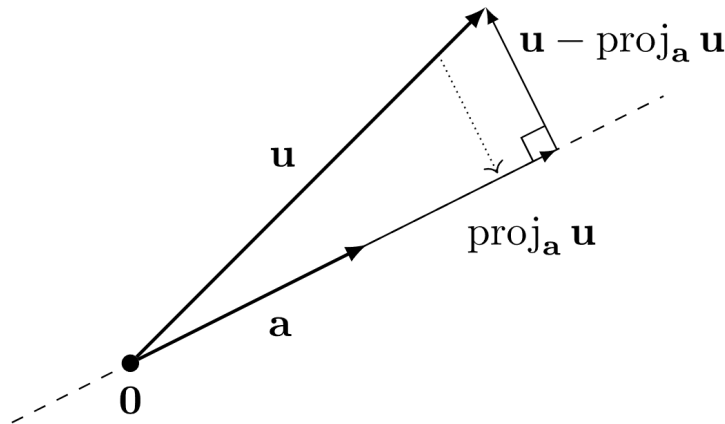
$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|. \quad (9)$$

- Know how to compute the orthogonal projection and what it geometrically represents

Definition: The **orthogonal projection** of $\mathbf{u}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$$

When the initial point of $\text{proj}_{\mathbf{a}} \mathbf{u}$ is placed at the origin, the terminal point will be the point closest to $\mathbf{u}$ through the origin and parallel to $\mathbf{a}$.



Tip: Vectors $\mathbf{a}$ and $\mathbf{u} - \text{proj}_{\mathbf{a}}\mathbf{u}$ are orthogonal.

Problem: Find the orthogonal projection of $\mathbf{u}$ onto $\mathbf{a}$ if $\mathbf{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and...

i. $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

ii. $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

iii. $\mathbf{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Solution: For (i): $\mathbf{u} \cdot \mathbf{a} = 1(3) + 1(0) = 3$ and $\|\mathbf{a}\|^2 = 3^2 + 0^2 = 9$, so $\text{proj}_{\mathbf{a}}\mathbf{u} = \frac{3}{9} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. If you draw this out, this makes sense; because $\mathbf{a}$ lies along the x -axis, we're just projecting the vector down to its x -coordinate.

For (ii): $\mathbf{u} \cdot \mathbf{a} = 2(3) + 0(0) = 6$ and $\|\mathbf{a}\|^2 = 3^2 + 0^2 = 9$, so $\text{proj}_{\mathbf{a}}\mathbf{u} = \frac{6}{9} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. If you draw this out, this makes sense; because $\mathbf{u}$ already runs along the same line as $\mathbf{a}$, $\mathbf{u}$ is its own projection.

For (iii): $\mathbf{u} \cdot \mathbf{a} = 0(3) + 1(0) = 0$ and $\|\mathbf{a}\|^2 = 3^2 + 0^2 = 9$, so $\text{proj}_{\mathbf{a}}\mathbf{u} = \frac{0}{9} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Because $\mathbf{u}$ and $\mathbf{a}$ are already orthogonal, when I project $\mathbf{u}$ onto $\mathbf{a}$, I get a zero point (i.e., the zero vector).

Problem: Find the orthogonal projection of $\mathbf{a}$ onto $\mathbf{u}$ if $\mathbf{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. (note the order is swapped!)

Solution: $\mathbf{a} \cdot \mathbf{u} = (3)(1) + (0)(1) = 3$ and $\|\mathbf{u}\|^2 = (1)^2 + (1)^2 = 2$. Now we have $\text{proj}_{\mathbf{u}}\mathbf{a} = \frac{3}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$.

- If given one of the 10 vector space properties, be able to verify if the property holds/does not hold for a potential vector space

Tip: Recall the following definition of vector spaces.

A real vector space is a set V of elements on which we have two operations, vector addition and scalar multiplication, defined with the following properties:

- (a) If $\mathbf{u}$ and $\mathbf{v}$ are any elements in V , then $\mathbf{u} + \mathbf{v}$ is in V . In this case, we say V is closed under addition.
 - (b) If $\mathbf{u}$ is any element in V and c is any real number, then $c\mathbf{u}$ is in V . In this case, we say V is closed under scalar multiplication.
1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .
 2. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
 3. There exists a zero vector $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ for any $\mathbf{u}$ in V . $\mathbf{0}$ is called the zero vector for V , and is sometimes denoted $\mathbf{0}_V$.
 4. For each $\mathbf{u}$ in V there exists an element $-\mathbf{u}$ in V such that $\mathbf{u} + -\mathbf{u} = -\mathbf{u} + \mathbf{u} = \mathbf{0}$. $-\mathbf{u}$ is called the negative of $\mathbf{u}$ or the additive inverse of $\mathbf{u}$.
 5. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ for any $\mathbf{u}, \mathbf{v}$ in V and any c in $\mathbb{R}$.
 6. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ for any $\mathbf{u}$ in V and c, d in $\mathbb{R}$.
 7. $c(d\mathbf{u}) = (cd)\mathbf{u}$ for any $\mathbf{u}$ in V and any c, d in $\mathbb{R}$.
 8. $1\mathbf{u} = \mathbf{u}$ for any $\mathbf{u}$ in V .

Warning: It may **NOT** be the case that $\mathbf{0}$ is the column vector whose entries are all equal to zero. $\mathbf{0}$ depends on the set V and the definition of vector addition (note that for “nonstandard vector addition,” we use $\oplus$, and for “nonstandard scalar multiplication,” we use $\odot$).

Theorem: Let V be a vector space. For any u in V and c in $\mathbb{R}$, then

1. $0\mathbf{u} = \mathbf{0}$
2. $c\mathbf{0} = \mathbf{0}$.
3. If $c\mathbf{u} = \mathbf{0}$ then $c = 0$ or $\mathbf{u} = \mathbf{0}$.
4. $(-1)\mathbf{u} = -\mathbf{u}$.

Tip: Recall the following common vector spaces:

Example: The set $\mathbb{R}^n$

Example: The set of all $m \times n$ matrices with real entries with matrix addition as $\oplus$ and scalar multiplication as $\odot$. This space is denoted as $M_{m \times n}$.

Example: The set of polynomials (in x) of degree at most d with the usual addition and scalar multiplication for polynomials as $\oplus$ and $\odot$. This space is denoted as P_d .

One can check that all the addition and scalar multiplication satisfies the vector properties outlined above.

Problem: Let V be the set of all positive real numbers; define $\oplus$ by $u \oplus v = uv$ ($\oplus$ is ordinary multiplication) and define $\odot$ by $c \odot v = v^c$. Prove that the zero vector $\mathbf{0}$ exists, and for each vector $\mathbf{v}$, find the negative of the vector, called $-\mathbf{v}$.

Solution: Using the properties of vector spaces, the zero vector $\mathbf{0}$ satisfies

$$\mathbf{u} \oplus \mathbf{0} = \mathbf{u} = \mathbf{0} \oplus \mathbf{u}. \quad (10)$$

We can rewrite the LHS of the equation to obtain

$$u \cdot \mathbf{0} = u. \quad (11)$$

By dividing both sides by u (because u is positive), we obtain that $\mathbf{0} = 1$. We still need to check that $\mathbf{u} = \mathbf{0} \oplus \mathbf{u}$. Indeed, we have

$$u = 1 \cdot u. \quad (12)$$

The negative of the vector $-\mathbf{v}$ satisfies

$$\mathbf{v} \oplus -\mathbf{v} = \mathbf{0} = -\mathbf{v} \oplus \mathbf{v}. \quad (13)$$

The equation simplifies to

$$v \cdot -\mathbf{v} = 1. \quad (14)$$

Therefore, $-\mathbf{v} = \frac{1}{v} = v^{-1}$. Indeed, we check that $-\mathbf{v} = -1 \odot \mathbf{v}$. Let's check that $\mathbf{0} = -\mathbf{v} \oplus \mathbf{v}$. This becomes

$$1 = \frac{1}{v} \cdot v. \quad (15)$$

- Be able to come up with a counterexample to show a set/operation is NOT a vector space

Problem: Let V be the set of all 2×1 matrices of form $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ with integer entries such that $|v_1 + v_2|$ is divisible by 5. Let the operation $\oplus$ be standard addition of matrices and the operation $\odot$ be standard scalar multiplication of matrices. Show that V is not a vector space.

Solution: We solve this problem by showing that V is not closed under scalar multiplication. consider the element $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$, which satisfies the condition of V : $|5 + 10| = 15 = (3)(5)$. Take the scalar $c = \frac{1}{5}$. Then the element $\frac{1}{5} \odot \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ does not belong to V , since this vector does not satisfy the condition that $1 + 2 = 3$ is divisible by 5. Hence, V is not closed under scalar multiplication, so it is not a vector space.

- Be able to verify that a subset of a vector space is or is NOT a subspace

Definition: Let V be a vector space and W be a nonempty subset of V . If W is a vector space with respect to the operations in V , then W is called a subspace of V .

Theorem: (Subspace Test) Let V be a vector space with vector addition and scalar multiplication. Let $W \subset V$ with W a nonempty subset of V . Then W is a subspace of V if and only if the following conditions hold.

1. If $\mathbf{u}$ and $\mathbf{v}$ are any vectors in W , then $\mathbf{u} + \mathbf{v}$ is in W .
2. If c is any real number and $\mathbf{u}$ is any vector in W , then $c\mathbf{u}$ is in W .

Problem: Prove that the set W of all 2×2 matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that $\det(A) = 0$ is not a subspace of $M_{2 \times 2}$.

Solution: We will show that for two elements $\mathbf{u}, \mathbf{v}$ in W , the element $\mathbf{u} + \mathbf{v}$ is not in W . Choose the following two matrices

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \quad (16)$$

But one can check that $A + B = I_2$, which has determinant 1.

Example: The set of all upper triangular matrices of $n \times n$ matrix is a subspace of $M_{n \times n}$.

Example: The set of all symmetric matrices of $n \times n$ matrix is a subspace of $M_{n \times n}$.

Problem: Prove that the set

$$H := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid x + y + z = 1 \right\} \quad (17)$$

is **not** a subspace of $\mathbb{R}^3$ with the usual addition and scalar multiplication.

Solution: We will show that if $\mathbf{u}$ is any vector in H , then a scalar multiplication of $\mathbf{u}$ may not lie in H . Indeed, consider $\mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Then the scalar multiple $2\mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$ is not in W because $0 + 0 + 2 = 2 \neq 1$.

Problem: Consider the set P_* of P_2 defined as

$$P_* := \{a_2x^2 + a_1x + a_0 \mid a_0 = 0\} \quad (18)$$

Prove that P_* is a vector space under the usual polynomial addition and scalar multiplication.

Solution: It suffices to show that P_* is a subspace of P_2 .

1. Nonempty: The element $0 = 0x^2 + 0x + 0$ is in P_*

2. Closed under addition: Note that if $a_2x^2 + a_1x + a_0$ and $b_2x^2 + b_1x + b_0$ are elements of P_* , then

$$(a_2x^2 + a_1x + a_0) + (b_2x^2 + b_1x + b_0) = (a_2 + b_2)x^2 + (a_1 + b_1)x + (a_0 + b_0) \quad (19)$$

Because $a_0 + b_0 = 0 + 0 = 0$, it holds that the addition of two vectors is also in P_* .

3. Closed under scalar multiplication: Note that if c is any constant, then

$$c(a_2x^2 + a_1x + a_0) = (ca_2)x^2 + (ca_1)x + (ca_0) \quad (20)$$

and $ca_0 = c \cdot 0 = 0$. Hence, it holds that the scalar multiple of a vector in P_* is also in P_* .

Thus, P_* is a subspace of P_2 , which shows that P_* is a vector space.

Problem: Consider P_2 and the subset of vectors $U = \{p(x) \text{ in } P_2 : p'(1) = 0\}$. Is U a subspace of P_2 ?

Solution: We use the subspace test to check:

1. Nonempty: If $p(x) = 0$ (i.e. the zero polynomial), then $p'(x) = 0$, so $p'(1) = 0$, so 0 is in U .
2. Closed under addition: Let $p(x)$ and $q(x)$ be in U : this means $p'(1) = 0, q'(1) = 0$. Then $(p + q)'(x) = p'(x) + q'(x)$, so $(p + q)'(1) = p'(1) + q'(1) = 0 + 0 = 0$. Thus, $p(x) + q(x)$ is also in U .
3. Closed under scalar multiplication: Let c be any real number, and let $p(x)$ be in U . Then $(cp)'(1) = cp'(1) = c(0) = 0$, so $cp(x)$ is also in U .

U satisfies the three conditions of the subspace test, so it is a subspace of P_2 .

- Be able to determine if a vector is in the span of a set of vectors

Definition: Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be vectors in a vector space V . A vector $\mathbf{v}$ in V is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$ if $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \sum_{i=1}^k a_i\mathbf{v}_i$ for some $a_1, \dots, a_k$ in $\mathbb{R}$.

Definition: If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a set of vectors in vector space V , then the set of all vectors in V that are linear combinations of the vectors in S is denoted by $\text{span}S$ or $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

Theorem: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a set of vectors in vector space V . Then $\text{span}S$ is a subspace of V .

Example: The vector $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ is **not** in the span of $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} \right\}$.

Example: The vector $\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ is in the span of $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} \right\}$.

- Be able to determine if a set spans a vector space

Definition: Let S be a set of vectors in a vector space V . If every vector in V is a linear combination of the vectors in S , then S **spans** V , or V **is spanned by** S . (i.e. $\text{span}S = V$).

Problem: Prove that the set $S := \{x + 3, x - 2\}$ spans the vector space P_1 .

Solution 1: To see this, notice that

$$1 = \frac{1}{5}(x + 3) - \frac{1}{5}(x - 2) \quad (21)$$

$$x = \frac{2}{5}(x + 3) + \frac{3}{5}(x - 2) \quad (22)$$

Hence, any element $ax + b$ in P_1 can be expressed as

$$ax + b = a\left(\frac{2}{5}(x+3) + \frac{3}{5}(x-2)\right) + b\left(\frac{1}{5}(x+3) - \frac{1}{5}(x-2)\right) = \left(\frac{2}{5}a + \frac{1}{5}b\right)(x+3) + \left(a\frac{3}{5} - b\frac{1}{5}\right)(x-2). \quad (23)$$

Because every element in P_1 is a linear combination of elements in S , we have that S spans P_1 .

Solution 2: Suppose $ax + b$ in P_1 is any element. Then we want to find a_1, a_2 in $\mathbb{R}$ such that

$$a_1(x+3) + a_2(x-2) = ax + b. \quad (24)$$

We can simplify the above equation to

$$(a_1 + a_2)x + (3a_1 - 2a_2) = ax + b \quad (25)$$

This is associated to the augmented matrix

$$\left[\begin{array}{cc|c} 1 & 1 & a \\ 3 & -2 & b \end{array} \right] \quad (26)$$

The RREF of the above matrix is

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{2}{5}a + \frac{1}{5}b \\ 0 & 1 & -\frac{1}{5}b + \frac{3}{5}a \end{array} \right] \quad (27)$$

Because the solution exists for a_1, a_2 for any choice of a, b , it holds that S spans P_1 .

Problem: Consider $W = \left\{ \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} \mid a_1 - 2a_2 + 3a_4 = 0, a_3 = 4a_2 \right\}$. Show W is a subspace of $\mathbb{R}^4$, and find a spanning set for W .

Solution: We leave as an exercise to justify W is a subspace of $\mathbb{R}^4$. To find a spanning set, note

that every vector in W is of the form $\begin{bmatrix} 2a_2 - 3a_4 \\ a_2 \\ 4a_2 \\ a_4 \end{bmatrix} = a_2 \begin{bmatrix} 2 \\ 1 \\ 4 \\ 0 \end{bmatrix} + a_4 \begin{bmatrix} -3 \\ 0 \\ 0 \\ 1 \end{bmatrix}$. Since every vector in W is

a linear combination of $\begin{bmatrix} 2 \\ 1 \\ 4 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, $W = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

- Be able to determine if two spanning sets generate the same subspace

Tip: Suppose S and T are two sets of vectors in a vector space. If each vector in S can be expressed as a linear combination of the vectors in T , and each vector in T can be expressed as a linear combination of the vectors in S , then $\text{span}S$ and $\text{span}T$ are the same space.

Example: Show that the span of $S = \{x^2 + 4, x^2 - 3\}$ and the span of $T = \{1, x^2\}$ are the same subspace of P_2 .

Solution: First, let's show that every vector in S is a linear combination of the vectors in T :

$$x^2 + 4 = 4(1) + 1(x^2)$$

$$x^2 - 3 = -3(1) + 1(x^2)$$

Now we show that every vector in T is a linear combination of the vectors in S :

$$1 = \frac{1}{7}(x^2 + 4) - \frac{1}{7}(x^2 - 3)$$

$$x^2 = \frac{3}{7}(x^2 + 4) + \frac{4}{7}(x^2 - 3)$$

Thus, these are both spanning sets for the same subspace of P_2 : It's easy to see that from T , this subspace is $\{ax^2 + bx + c \text{ in } P_2 \mid b = 0\}$.

- Be able to determine if a set is linearly independent/linearly dependent (via systems of equations, the determinant of a matrix, or demonstrating a linear combination)

Definition: The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in a vector space V are said to be **linearly dependent** if there exists $a_1, \dots, a_k$ real numbers such that

$$\sum_{i=1}^k a_i \mathbf{v}_i = a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = \mathbf{0}.$$

and not all $a_i = 0$.

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in a vector space V are said to be **linearly independent** if whenever $\sum_{i=1}^k a_i \mathbf{v}_i = \mathbf{0}$, then $a_1 = a_2 = \dots = a_n = 0$.

Tip: To show that a set of n vectors in $\mathbb{R}^n$ is linearly independent, it suffices to show that the determinant of the matrix (obtained by putting the n vectors as columns of the matrix) is not equal to 0. The determinant is equal to 0 if and only if the set is linearly dependent.

Problem: For what values of c is $S = \{x^2 + x - 1, (c^2 - 4)x, x^2 - 3\}$ linearly independent?

Solution: We wish to see when the linear combination

$$a_1(x^2 + x - 1) + a_2((c^2 - 4)x) + a_3(x^2 - 3) = 0 = 0x^2 + 0x + 0$$

has a nontrivial solution (this would mean the set is NOT linearly independent).

This is equivalent to solving the homogeneous system $a_1 + a_3 = 0, a_1 + (c^2 - 4)a_2 = 0, -a_1 - 3a_3 = 0$.

It suffices to check that the coefficient matrix of this homogeneous system has determinant not equal to zero:

$$\det \begin{bmatrix} 1 & 0 & 1 \\ 1 & c^2 - 4 & 0 \\ -1 & 0 & -3 \end{bmatrix} = -2(c^2 - 4) \neq 0. \quad (28)$$

Hence, the set is linearly independent if and only if $c^2 - 4 \neq 0$, or equivalently $c \neq \pm 2$.

- Know how to show a set is or is NOT a basis for a vector space

Definition: The vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V are said to form a **basis** for V if

1. $\mathbf{v}_1, \dots, \mathbf{v}_k$ span V , and
2. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent.

Theorem: If S is a basis for a vector space V , then every vector in V can be written in one and only one way as a linear combination of the vectors in S .

Tip: To show whether a set S is a basis for a vector space V , we need to show:

1. $\text{Span} S = V$
2. S is linearly independent.

Example: The set $S := \{x + 3, x - 2\}$ is in fact a basis of the vector space P_1 .

Problem: Find a basis for $U = \{p(x) \text{ in } P_2 : p'(1) = 0\}$.

Solution: Let's start by finding a spanning set for U . If $p(x) = ax^2 + bx + c$ is in U , then $p'(1) = 0$. $p'(x) = 2ax + b$, so we substitute in $x = 1$: $p'(1) = 2a + b = 0$. This gives us a relationship between the coefficients in $p(x)$: we need $b = -2a$, so $p(x)$ must have the form $p(x) = ax^2 - 2ax + c = a(x^2 - 2x) + c(1)$. A spanning set for U is $\{x^2 - 2x, 1\}$. To show that it is a basis, we need to

demonstrate that the set is linearly independent. We set up an arbitrary linear combination and set it equal to zero:

$$a_1(x^2 - 2x) + a_2(1) = 0 \Rightarrow a_1x^2 - 2a_1x + a_2 = 0.$$

Each term of the polynomial must be zero, so $a_1 = 0$ and $a_2 = 0$, so this set is linearly independent. This means $\{x^2 - 2x, 1\}$ is a basis for U .

- Know how to find a basis from a set of vectors

Theorem: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a set of nonzero vectors in a vector space V , and let $W = \text{Span}S$. Then some subset of S is a basis of W .

Tip: In the case that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ with $\text{Span}S = W \subset \mathbb{R}^m$, we can find a subset of S that is a basis of W in the following way.

1. Form the equation $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{0}$.
2. Construct the coefficient matrix A associated with the homogeneous system above (i.e., the i th column of A is the vector $\mathbf{v}_i$). **Transform the matrix A into RREF.**
3. The column vectors of A corresponding to columns containing **leading ones (pivots)** in the $\text{RREF}(A)$ form a basis for W .

Problem: Find a basis for $\mathbb{R}^4$ that contains $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 4 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \\ 0 \\ 2 \end{bmatrix}$.

Solution: Note that $\mathbf{u}$ and $\mathbf{v}$ are linearly independent. We need to find two additional vectors to form a basis for $\mathbb{R}^4$.

We notice that if we include the four standard basis vectors for $\mathbb{R}^4$ with $\mathbf{u}$ and $\mathbf{v}$, this will automatically form a spanning set for $\mathbb{R}^4$, so to find a basis, we set up the augmented matrix with $\mathbf{u}$ and $\mathbf{v}$ first:

$$\begin{bmatrix} 3 & -1 & 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 4 & 2 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -1/6 & 0 & 1/3 \\ 0 & 1 & 0 & 1/3 & 0 & -1/6 \\ 0 & 0 & 1 & 5/6 & 0 & -7/6 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Columns 1, 2, 3 and 5 contain leading ones, so we use the corresponding columns from the original

matrix: $\begin{bmatrix} 3 \\ 2 \\ 0 \\ 4 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 4 \\ 0 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ form a basis for $\mathbb{R}^4$.

- Be able to find the coordinates of a vector with respect to an ordered basis

Definition: Given an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ for a vector space V and a vector $\mathbf{v}$ in V ,

the **coordinate vector** for $\mathbf{v}$ relative to the basis B is $C_B(\mathbf{v}) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ if and only if

$$\mathbf{v} = a_1 \mathbf{b}_1 + a_2 \mathbf{b}_2 + \cdots + a_n \mathbf{b}_n.$$

Example: Given an ordered basis $B = (x^2 + 2, x + 1, 3)$ of P_2 (note: verify B is a basis!), the polynomial $p = x^2 + x + 2$ can be written as

$$x^2 + x + 2 = 1 \cdot (x^2 + 2) + 1 \cdot (x + 1) + \left(-\frac{1}{3}\right) \cdot 3. \quad (29)$$

Hence, as a column vector, we can write p as

$$C_B(p) = \begin{bmatrix} 1 \\ 1 \\ -\frac{1}{3} \end{bmatrix}. \quad (30)$$

Problem: Consider $V = P_2$ with $B = (x + 2, x^2, 1)$ (note: verify B is a basis!). Find p if $C_B(p) = \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix}$.

Solution: If $C_B(p) = \begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix}$, then following the order of B , 4 is the coefficient of $x + 2$, -2 is the coefficient of x^2 , and 5 is the coefficient of 1, so $p = 4(x + 2) - 2x^2 + 5 = -2x^2 + 4x + 13$.

Problem: Let V = the subspace of 2×2 symmetric matrices (which is itself a vector space), and $B = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$. Find $C_B(M)$ for $M = \begin{bmatrix} -3 & 4 \\ 4 & 6 \end{bmatrix}$.

Solution: $M = \begin{bmatrix} -3 & 4 \\ 4 & 6 \end{bmatrix} = -3 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 4 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + 6 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, so $C_B(M) = \begin{bmatrix} -3 \\ 4 \\ 6 \end{bmatrix}$.

- Be able to find the dimension of a vector space.

Definition: The dimension of a vector space V is the number of vectors in a basis of V .

Problem: What's the dimension of $U = \{p(x) \text{ in } P_2 : p'(1) = 0\}$?

Solution: We found a basis for U earlier: $\{x^2 - 2x, 1\}$. Since there are two vectors in this basis, the dimension of U is 2.

- Know the dimensions of the common finite-dimensional vector spaces: $\mathbb{R}^n, P_d, M_{m \times n}$

Example: The dimension of $\mathbb{R}^n$ is equal to n , with standard basis given as

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right\} \quad (31)$$

Example: The dimension of $M_{m \times n}$ is equal to mn (multiplication of m and n). One possible basis (sometimes called a “standard basis”) of $M_{2 \times 3}$ is given as

$$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \quad (32)$$

Example: The dimension of P_d , the set of polynomials of degree at most d , is equal to $d + 1$, with standard basis given as

$$\{1, x, x^2, \dots, x^d\} \quad (33)$$

- Know how to use the dimension of a vector space to justify if a set is a basis

Tip: If $\dim V = n$, and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is set of vectors in V , then:

- a. If S spans V , then S is a basis for V (i.e. S will then be linearly independent).
- b. If S is linearly independent, then S is a basis for V (i.e. S will then span V).

Example: We can check that the set $S = \{x^2 + 2, x^2 - 3, 2x + 1\}$ is a basis of P_2 just showing that the set of vectors is linearly independent (check this!) and the number of elements in S is equal to 3, which is the dimension of P_2 .

- Be able to find a basis for the row space, column space/image space, and nullspace of a matrix

Tip: A basis for the row space is the set of nonzero row vectors in the $RREF(A)$.

Definition: The image of a matrix A , $\text{im } A$ is the set of vectors $\mathbf{b}$, where $\mathbf{b} = A\mathbf{x}$ for some vector $\mathbf{x}$.

Note: Since $A\mathbf{x}$ is a linear combination of the columns of A (with the values in $\mathbf{x}$ the scalars), this means that the columns of A form a spanning set for $\text{im } A$. This means that the column space of A and the image of A are EXACTLY the same subspace.

Tip: We can follow the following steps to find a basis for the image of a matrix A . This process is called “pick up the pivots!”

1. Find the RREF of A .
2. Identify the columns in $RREF(A)$ with leading ones/pivots.
3. The corresponding columns in A form a basis for $\text{im } A$ (and the column space of A).

Example: Suppose A is the following matrix

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 3 \\ 2 & 1 & 0 \\ 2 & -3 & 8 \end{bmatrix} \quad (34)$$

and the RREF of A is given by

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (35)$$

The nonzero rows of the RREF of A are a basis for the row space: $[1 \ 0 \ 1]$ and $[0 \ 1 \ -2]$.

The first and the second columns of RREF of A contains the leading ones. Hence, the first and the

second column vector of A spans the column space/image space of A : $\begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \\ 1 \\ -3 \end{bmatrix}$.

Tip: We can follow the following steps to find a basis for the null space of a matrix.

1. Find the RREF of A .
2. Using the RREF of A , use the free variables to express a general form of solution to the equation $A\mathbf{x} = \mathbf{0}$.
3. Pull apart the general solution to express the vectors in the nullspace of A as a linear combination of explicit vectors. These explicit vectors are the basis for the nullspace of A .

Example: A and the RREF of A are given below:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 3 \\ 2 & 1 & 0 \\ 2 & -3 & 8 \end{bmatrix} \quad RREF(A) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (36)$$

Then with regards to the system of equations $A\mathbf{x} = \mathbf{0}$ with $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, we can pick x_3 as the free variable $x_3 = t$. The RREF implies $x_1 = -t$ and $x_2 = 2t$. Hence, every solution for $A\mathbf{x} = \mathbf{0}$ can be written as

$$\begin{bmatrix} -t \\ 2t \\ t \end{bmatrix} = t \cdot \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad (37)$$

This implies that the vector $\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ spans the nullspace of A (and is in fact a basis for $\text{null}(A)$).

- Be able to compute the dimension of the row space and the dimension of the column space of a matrix

Tip: the dimension of the row space and the dimension of the column space of a matrix A are equal, called the **rank** of the matrix A . This can be computed from the **number of leading ones (pivots)** in the RREF of A .

Example: Suppose the RREF of a 4×3 matrix A is given by

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (38)$$

Then the dimension of the row space and the dimension of the column space of A are both 2, the number of leading ones.

- Be able to find the rank and nullity for a matrix, and how the nullity and rank relate to the matrix

Definition: The nullity of A is the dimension of the null space of A (i.e. the set of all solutions to $A\mathbf{x} = \mathbf{0}$.)

Theorem: (Rank-Nullity Theorem)

$$\text{Rank}(A) + \dim \text{null}(A) = \# \text{ of columns of } A \quad (39)$$

In particular, $\dim \text{null}(A) = \text{Nullity}(A)$ is the number of free variables determining the set of solutions to $A\mathbf{x} = \mathbf{0}$, and $\text{Rank}(A)$ is the number of leading ones / pivots of the RREF of A .

Example: Suppose the REF of A is given by

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (40)$$

Then the equation $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions with 1 free variable. This implies that $\text{Nullity}(A) = \dim \text{null}(A) = 1$. Furthermore, we have $\text{Rank}(A) = 2$ because the number of leading ones is equal to 2. Hence, the rank-nullity theorem holds, because $2 + 1 = 3$.

- Know how to compute the change matrix/transition matrix given two ordered bases for a vector space, and how to use it to find the coordinate vector

Definition: Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ and D be two ordered basis of an n -dimensional real vector space V .

Then the change matrix/transition matrix $P_{D \leftarrow B}$ from the B -basis to the D -basis is given by

$$P_{D \leftarrow B} = [C_D(\mathbf{b}_1) \ C_D(\mathbf{b}_2) \ \cdots \ C_D(\mathbf{b}_n)] \quad (41)$$

Tip: We in fact have the relation

$$C_D(\mathbf{v}) = P_{D \leftarrow B} C_B(\mathbf{v}) \quad (42)$$

Problem: If $B = (1, x)$ and $D = (x + 5, x - 4)$, build the transition matrix $P_{D \leftarrow B}$ and use it to compute $C_D(p)$ for $p = 4x + 7$.

Solution: To obtain the transition matrix $P_{D \leftarrow B}$, we need to represent the basis vectors of B with respect to D . We can solve $1 = a_1(x + 5) + a_2(x - 4)$ to find that $a_1 = 1/9, a_2 = -1/9$, and we can solve $x = c_1(x + 5) + c_2(x - 4)$ to find that $c_1 = 4/9, c_2 = 5/9$, so

$$P_{D \leftarrow B} = \begin{bmatrix} 1/9 & 4/9 \\ -1/9 & 5/9 \end{bmatrix}$$

Now we compute $C_D(p) = P_{D \leftarrow B} C_B(p)$ for $p = 4x + 7$: since $C_B(p) = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$, this is

$$C_D(p) = \begin{bmatrix} 1/9 & 4/9 \\ -1/9 & 5/9 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 23/9 \\ 13/9 \end{bmatrix}$$

Let's verify this: If $C_D(p) = \begin{bmatrix} 23/9 \\ 13/9 \end{bmatrix}$, then $p = 23/9(x + 5) + 13/9(x - 4)$:

$$\frac{23}{9}(x + 5) + \frac{13}{9}(x - 4) = \frac{23 + 13}{9}x + \frac{23 \cdot 5 - 13 \cdot 4}{9} = 4x + 7 = p.$$

Problem: Let $B = (1, x)$ and $D = (x, 1)$ be the ordered basis of P_1 . Find the transition matrix $P_{D \leftarrow B}$.

Solution: To obtain the transition matrix $P_{D \leftarrow B}$, we need to represent the basis vectors of B with respect to D . Note that

$$1 = 0 \cdot x + 1 \cdot 1. \quad (43)$$

Hence, $C_D(1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Likewise, we have $C_D(x) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Hence, the transition matrix $P_{D \leftarrow B}$ is obtained by concatenating these two column vectors:

$$P_{D \leftarrow B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (44)$$

Problem: Let $B = (p_1, p_2)$ be an unknown ordered basis for P_1 , and let $D = (2x + 1, 3x - 4)$ be an ordered basis for P_1 . If $P_{D \leftarrow B} = \begin{bmatrix} -4 & 2 \\ 1 & 0 \end{bmatrix}$ is the transition matrix from B to D , find p_1 and p_2 .

Solution: The first column of $P_{D \leftarrow B}$ is $C_D(p_1)$, and the second column of $P_{D \leftarrow B}$ is $C_D(p_2)$, so $C_D(p_1) = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ and $C_D(p_2) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Since we know D , we can write out p_1 and p_2 :

$$\begin{aligned} p_1 &= -4(2x + 1) + 1(3x - 2) \\ p_2 &= 2(2x + 1) + 0(3x - 4) \end{aligned}$$

- Know the basic properties of change matrices/transition matrices (composition, inversion)

Tip: Let B , D , and E denote ordered bases of a vector space V . $P_{E \leftarrow D} P_{D \leftarrow B} = P_{E \leftarrow B}$.

Tip: By construction, we have

$$P_{D \leftarrow B}^{-1} = P_{B \leftarrow D} \quad (45)$$

In particular, every change matrix is invertible.

Problem: Suppose $B = (4x - 1, x)$, $D = (1, x)$, and $E = (3, x + 5)$. Use the fact that $P_{E \leftarrow D} P_{D \leftarrow B} = P_{E \leftarrow B}$ to find $P_{E \leftarrow B}$.

Solution: $P_{D \leftarrow B}$ is easy to compute: We just write each element in B using its coordinate representation in $D = (1, x)$, so

$$P_{D \leftarrow B} = \begin{bmatrix} -1 & 0 \\ 4 & 1 \end{bmatrix}.$$

$P_{E \leftarrow D}$ requires a little more work: $1 = \frac{1}{3}(3) + 0(x + 5)$, and $x = -\frac{5}{3}(3) + 1(x + 5)$, so $P_{E \leftarrow D} = \begin{bmatrix} 1/3 & -5/3 \\ 0 & 1 \end{bmatrix}$

$$\text{Now } P_{E \leftarrow B} = P_{E \leftarrow D} P_{D \leftarrow B} = \begin{bmatrix} 1/3 & -5/3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} -7 & -5/3 \\ 4 & 1 \end{bmatrix}$$

Is this correct? Let's check! It should be true that $C_E(4x - 1)$ is the first column, and $C_E(x)$ is the second column:

$$-7(3) + 4(x + 5) = 4x + 20 - 21 = 4x - 1$$

and

$$-\frac{5}{3}(3) + 1(x + 5) = x + 5 - 5 = x.$$

So this matrix is correct!

Problem: Let $B = (x + 3, 2 - x)$ and $D = (\mathbf{w}_1, \mathbf{w}_2)$ be ordered bases of P_1 . If the transition matrix $P_{D \leftarrow B}$ is given as $P_{D \leftarrow B} = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$, find D .

Solution: Note that we have the relation

$$C_D(v) = P_{D \leftarrow B} C_B(v) \quad (46)$$

We plug in $v = \mathbf{w}_1$ to obtain that $C_D(\mathbf{w}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} C_B(\mathbf{w}_1). \quad (47)$$

The goal is to figure out $C_B(\mathbf{w}_1)$. That way, we can obtain in explicit polynomial form what the vector $\mathbf{w}_1$ is. The inverse of the matrix can be computed as

$$P_{D \leftarrow B}^{-1} = P_{B \leftarrow D} = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{3}{5} \end{bmatrix} \quad (48)$$

We multiply both sides of the equation by $P_{D \leftarrow B}^{-1}$ to the left to obtain

$$C_B(\mathbf{w}_1) = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} \\ \frac{1}{5} \end{bmatrix}. \quad (49)$$

which implies

$$\mathbf{w}_1 = \frac{1}{5}(x+3) + \frac{1}{5}(2-x) = 1. \quad (50)$$

Likewise, we plug in $v = \mathbf{w}_2$ to obtain

$$C_B(\mathbf{w}_2) = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} \\ \frac{3}{5} \end{bmatrix} \quad (51)$$

which implies

$$\mathbf{w}_2 = -\frac{2}{5}(x+3) + \frac{3}{5}(2-x) = -x. \quad (52)$$

Hence, we have $D = (1, -x)$.

- Know how to verify if a vector is an eigenvector for a matrix.

Tip: For a square $(n \times n)$ matrix A , a *non-zero* vector $\mathbf{v}$ is an eigenvector for A if $A\mathbf{v} = \lambda\mathbf{v}$ for some scalar λ .

Warning: An eigenvector can NOT be the zero vector. However, λ can be zero.

For example, if $A = \begin{bmatrix} -2 & 3 \\ 4 & -6 \end{bmatrix}$, I can verify that $\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is an eigenvector for A :

$$A\mathbf{v} = \begin{bmatrix} -2 & 3 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \text{ so } \mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ is an eigenvector of } A \text{ with eigenvalue } \lambda = 0.$$

$\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ is another eigenvector for A . I'll leave it as an exercise to find the corresponding eigenvalue.

- Be able to compute the eigenvectors/eigenvalues (real eigenvalues only) for a matrix using the characteristic equation, and find a basis for the eigenspace

Tip: To find eigenvalues/eigenvectors for a matrix A , we wish to find λ and nonzero $\mathbf{v}$ that satisfy the equation $A\mathbf{v} = \lambda\mathbf{v}$. This is equivalent to solving $(\lambda I_n - A)\mathbf{v} = \mathbf{0}$, which will have a nonzero solution $\mathbf{v}$ exactly when $\det(\lambda I_n - A) = 0$.

Tip: Since $(\lambda I_n - A)\mathbf{v} = \mathbf{0}$ is a homogeneous system, the set $\{\mathbf{v} : A\mathbf{v} = \lambda\mathbf{v}\}$ forms a subspace of $\mathbb{R}^n$, called the eigenspace of A associated to λ . We denote this by $E_\lambda(A)$, i.e. $E_\lambda(A)$ is the eigenspace of A corresponding to λ .

The process is outlined below:

1. Find the solution of the **characteristic polynomial**

$$\det(\lambda I_n - A) = 0. \quad (53)$$

for the variable λ . If A is an $n \times n$ matrix, then the characteristic polynomial is a degree n polynomial of λ . The solutions of the equation above are **eigenvalues** of A .

2. Plug in each solution λ into $\lambda I_n - A$ to find the solutions to $(\lambda I_n - A)\mathbf{v} = \mathbf{0}$. The set of nonzero vectors $\mathbf{v}$ are **eigenvectors** of A with eigenvalue λ .
3. Since $(\lambda I_n - A)\mathbf{v} = \mathbf{0}$ is a homogeneous system, we can construct a basis for the solution set.

For example, if we want to find the eigenvalues and a basis for each eigenspace for $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$, then

$$\text{we solve } \det(\lambda I_2 - A): \det(\lambda I_2 - A) = \det \left(\begin{bmatrix} \lambda - 4 & 2 \\ -1 & \lambda - 1 \end{bmatrix} \right) = (\lambda - 4)(\lambda - 1) + 2 = \lambda^2 - 5\lambda + 6 = (\lambda - 3)(\lambda - 2) = 0$$

So our eigenvalues are $\lambda = 3, 2$.

Plugging in $\lambda = 3$, we get $\begin{bmatrix} 3-4 & 2 \\ -1 & 3-1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix}$, so we're solving $\begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Reducing $\begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix}$, we get $\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$, so $v_2 = t, v_1 = 2t$, and our eigenvectors for $\lambda = 3$ look like $\begin{bmatrix} 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ for nonzero t .

In particular, $\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace for $\lambda = 3$.

Plugging in $\lambda = 2$, we get $\begin{bmatrix} 2-4 & 2 \\ -1 & 2-1 \end{bmatrix} = \begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix}$, so we're solving $\begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Reducing $\begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix}$, we get $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, so $v_2 = t, v_1 = t$, and our eigenvectors for $\lambda = 2$ look like $\begin{bmatrix} t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for nonzero t .

In particular, $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace for $\lambda = 2$.

- Know how to use similarity to find shared properties of similar matrices

Definition: A is similar to B if $B = P^{-1}AP$ for some invertible matrix P .

Warning: Similarity is NOT the same thing as row-equivalent. $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is row-equivalent to $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, but they are not similar matrices.

Theorem: If A and B are similar matrices, then A and B have the same determinant, rank, trace, characteristic polynomial, and eigenvalues.

Recall: For an $n \times n$ matrix A , the trace of A , $\text{tr}(A)$, is the sum of the entries on the main diagonal.

Problem: Show that $A = \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ are NOT similar.

Solution: $\det(A) = 6 + 4 = 10$, while $\det(B) = 6 + 1 = 7$. Since $10 \neq 7$, these matrices are NOT similar.

Problem: Let $A = \begin{bmatrix} 4 & 1 \\ 2 & 2 \end{bmatrix}$, and $B = \begin{bmatrix} 0 & a \\ 3 & b \end{bmatrix}$. If A is similar to B , find a and b .

Solution: If A and B are similar, they have the same trace: $\text{tr}(A) = 4 + 2 = 6 = 0 + b = \text{tr}(B)$, so $b = 6$.

A and B also have the same determinant: $\det(A) = (4)(2) - 2 = 6 = (0)(6) - 3a = \det(B)$, so $a = -2$.

- Be able to find the algebraic multiplicity and geometric multiplicity of an eigenvalue

Definition: If λ_i is an eigenvalue of A , the algebraic multiplicity of λ_i is the highest power that the factor $(\lambda - \lambda_i)$ appears in the characteristic polynomial of A , i.e. if $c_A(\lambda) = (\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2} \cdots (\lambda - \lambda_k)^{m_k}$ is the characteristic polynomial of A , where $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct eigenvalues, then the algebraic multiplicity of λ_i is m_i .

The geometric multiplicity of an eigenvalue λ for a matrix A is $\dim E_\lambda(A)$, where $E_\lambda(A)$ is the eigenspace of A corresponding to λ .

Tip: The geometric multiplicity of an eigenvalue is always less than or equal to its algebraic multiplicity.

Problem: Consider the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 3 & -2 \\ 2 & 0 & 5 \end{bmatrix}$. Find the algebraic multiplicity of each eigenvalue, a basis for each eigenspace, and the geometric multiplicity of each eigenvalue.

Solution:

$$\text{Solve } 0 = \det(\lambda I_3 - A) = \det \begin{bmatrix} \lambda - 3 & 0 & 0 \\ 2 & \lambda - 3 & 2 \\ -2 & 0 & \lambda - 5 \end{bmatrix} = (\lambda - 3)(\lambda - 3)(\lambda - 5) = (\lambda - 3)^2(\lambda - 5).$$

So $\lambda = 3$ is an eigenvalue of A with algebraic multiplicity 2, and $\lambda = 5$ is an eigenvalue of A with algebraic multiplicity 1.

Plugging $\lambda = 5$ into $\lambda I_3 - A$ gives

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 2 \\ -2 & 0 & 0 \end{bmatrix}.$$

We can reduce this to its reduced row echelon form to find eigenvectors $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$:

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 2 \\ -2 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

This means for the solution: $a = 0$, $b = -r$, and $c = r$, so $\left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace of $\lambda = 5$.

Plugging in $\lambda = 3$ into $\lambda I_3 - A$ gives

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 2 \\ -2 & 0 & -2 \end{bmatrix}.$$

We can reduce this to its reduced row echelon form:

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 2 \\ -2 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This means: $a = -s$, $b = r$, and $c = s$, so our eigenvectors look like $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -s \\ r \\ s \end{bmatrix} = s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + r \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$

so $\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis for the eigenspace of $\lambda = 3$.

The geometric multiplicity of $\lambda = 5$ is 1, and the geometric multiplicity of $\lambda = 3$ is 2.

- Be able to determine if a matrix is diagonalizable

Definition: A matrix A is diagonalizable if it is similar to a diagonal matrix D , i.e. there exists an invertible matrix P with $P^{-1}AP = D$

Theorem: Let A be an $n \times n$ matrix. A is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ consisting of eigenvectors of A .

When this is the case, the matrix $P = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_n]$ is invertible and $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ where, for each i , λ_i is the eigenvalue of A corresponding to $\mathbf{x}_i$.

Fact: A couple useful facts about eigenvectors/diagonalizability:

- If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ of a matrix A , then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a linearly independent set.
- If A is an $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.
- A matrix with is diagonalizable if and only if each eigenvalue has geometric multiplicity equal to its algebraic multiplicity.

Problem: Suppose A is a 5×5 matrix with characteristic polynomial $c_A(\lambda) = \lambda(\lambda^2 - 4)(\lambda + 7)(\lambda - 1)$. Is this enough information to tell if A is diagonalizable?

Solution: Finishing factoring the polynomial, we get $c_A(\lambda) = \lambda(\lambda + 2)(\lambda - 2)(\lambda + 7)(\lambda - 1)$, so our eigenvalues are $\lambda = -2, 0, 1, 2, 7$. Since we have 5 distinct eigenvalues, and A is 5×5 , A is automatically diagonalizable.

Note: It's okay that $\lambda = 0$ is an eigenvalue of A . This actually means that A is NOT invertible (justify this!)

Problem: Is $A = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 3 & -2 \\ 2 & 0 & 5 \end{bmatrix}$ diagonalizable?

Solution: Yes! We found bases for its eigenspaces earlier:

$\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ for $\lambda = 3$, and $\left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$ for $\lambda = 5$.

$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ are linearly independent, so they form a basis for $\mathbb{R}^3$, so A is diagonalizable.

Problem: Is $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ diagonalizable?

Solution: $\det(\lambda I_3 - A) = \det \begin{bmatrix} \lambda - 1 & 2 & 3 \\ 0 & \lambda - 1 & 2 \\ 0 & 0 & \lambda - 1 \end{bmatrix} = (\lambda - 1)^3$, so $\lambda = 1$ is our only eigenvalue (with algebraic multiplicity 3).

Finding our eigenvectors:

$$(1)I_3 - A = \begin{bmatrix} 0 & -2 & -3 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix},$$

which reduces to $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Ruh-roh. I see only 1 free variable, so my eigenspace only has dimension 1.

In other words, if $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is an eigenvector of A , then $a = t$ (a free variable), and $b = c = 0$, so my basis

consists of only one vector: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. So I cannot find a basis for $\mathbb{R}^3$ made of eigenvectors of A , so A is NOT diagonalizable.

Another valid argument: $\lambda = 1$ has algebraic multiplicity 3 but geometric multiplicity 1 (based on our previous work). Since the multiplicities don't match, A is not diagonalizable.

- Be able to diagonalize a matrix (i.e. find P and D)

Tip: For a diagonalizable $n \times n$ matrix A , we form P by using as its columns n linearly independent eigenvectors of A . Since the columns are linearly independent, P is invertible (Box of Facts!), so P^{-1} exists. $P^{-1}AP$ is a diagonal matrix with the eigenvalues of A along the diagonal.

Tip: The order you put the eigenvectors of A into P will decide the order of the eigenvalues in the matrix D .

Problem: Diagonalize $A = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 3 & -2 \\ 2 & 0 & 5 \end{bmatrix}$.

Solution: We found 3 linearly independent eigenvectors before: $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$, so we can use them for P .

$$P = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}, P^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \text{ and } P^{-1}AP = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} = D \text{ (as expected).}$$

- Use a diagonalization to compute higher powers of a matrix

Tip: For a diagonal matrix $D = \text{diag}(d_1, d_2, \dots, d_n)$, we can compute D^k just by just raising each entry to the k th power, i.e. $D^k = \text{diag}(d_1^k, d_2^k, \dots, d_n^k)$.

Problem: Compute A^8 for $A = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 3 & -2 \\ 2 & 0 & 5 \end{bmatrix}$. Write it as a single 3×3 matrix, but you don't need to simplify the entries.

Solution: We found in the previous problem that for $P = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ and

$$P^{-1}AP = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} = D. \text{ This means that } A = PDP^{-1}, \text{ so}$$

$$A^8 = (PDP^{-1})^8 = (PDP^{-1})(PDP^{-1}) \dots (PDP^{-1}) = PD^8P^{-1}.$$

$$D^8 = \begin{bmatrix} 3^8 & 0 & 0 \\ 0 & 3^8 & 0 \\ 0 & 0 & 5^8 \end{bmatrix}, \text{ so now we just compute the matrix product:}$$

$$A^8 = PD^8P^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3^8 & 0 & 0 \\ 0 & 3^8 & 0 \\ 0 & 0 & 5^8 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3^8 & 0 & 0 \\ 3^8 - 5^8 & 3^8 & 3^8 - 5^8 \\ -3^8 + 5^8 & 0 & 5^8 \end{bmatrix}$$

• *** Be familiar with the (evergrowing) list of invertibility criteria (Box of Facts)

The following statements are equivalent.

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n .
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be expressed as a product of some number of elementary matrices.
7. $\det(A) \neq 0$.
8. The rows of A form a linearly independent set of n vectors in $\mathbb{R}^n$.
9. The columns of A form a linearly independent set of n vectors in $\mathbb{R}^n$.
10. The rows of A span $\mathbb{R}^n$.
11. The columns of A span $\mathbb{R}^n$.
12. The dimension of the nullspace of A is zero.
13. The dimension of the image space/column space of A is n .
14. The scalar $\lambda = 0$ is not an eigenvalue of A .